

# Hitesh Singh

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## Education

### Kalinga Institute of Industrial Technology

2024-2028

*Bachelor of Technology in Computer Science Engineering (AI & ML) : CGPA: 9.48/10.00 (No Active Backlogs) Bhubaneswar, Odisha*

**Relevant Coursework:** Data Structures and Algorithms (Grade: Excellent), Object-Oriented Programming (Java), Operating Systems (Linux), Database Management Systems, Computer Networks, Distributed Operating Systems, Software Engineering, Probability and Statistics, Discrete Mathematics, Machine Learning, Artificial Intelligence.

## Experience

### Microsoft Learning Student Ambassador KIIT —

Aug 2025 – Present

- Managed end-to-end logistics, marketing, and speaker coordination for a large-scale technical event with 3,000+ registrations, ensuring operational excellence and high attendee engagement.
- Led a collaborative team of 10 AI/ML enthusiasts using Agile/Scrum methodologies and GitHub for version control to design and implement 5 innovative solutions, including a specialized NLP-based tool.
- Facilitated technical operations for a Competitive Programming Hackathon with 500+ participants, fostering an inclusive and innovative culture through real-time troubleshooting and participant support.

## Projects

### Autonomous Multi-Agent Financial Research Engine | yfinance, Agno, JUnit, LanceDB, Python

Jan 2026 - Feb 2026

- Architected a modular and distributed multi-agent system utilizing a "Supervisor-Worker" model to automate end-to-end investment research, aligning with standards for large-scale and resilient technology solutions.
- Ensured high-quality software delivery by implementing a comprehensive suite of JUnit/Mockito tests for agent tool-calling logic, achieving robust Application Resiliency.
- Engineered a high-performance RAG pipeline using Big Data processing concepts to perform semantic search on 100+ page SEC filings, ensuring 100% data grounding and automated citation.
- Integrated real-time technical analysis tools via yfinance, enabling autonomous agents to execute Python code for calculating complex market indicators like RSI and Moving Averages.
- Developed a Model Context Protocol (MCP) server to standardize agent-tool interactions, ensuring deterministic, schema-validated execution across specialized financial agents.
- Optimized system performance and cost-efficiency by migrating to a Gemini-native ecosystem, achieving sub-5 second response latencies while maintaining end-to-end project ownership.

### PokiDesk: AI-Powered Multi-Modal Sign Language Interface | MediaPipe, Groq LLM API, Python, OpenCV

Dec 2025

- Developed the end-to-end development of a real-time communication tool utilizing MediaPipe for hand-gesture tracking, converting sign language into audible speech to promote an inclusive and innovative culture.
- Integrated Groq LLM to achieve low-latency processing, enabling near-instant translation of complex gestures while ensuring application resiliency during high-concurrency usage.
- Designed a deterministic pipeline for real-time translation, utilizing schema-validated technical processes to ensure 98% accuracy in gesture-to-text mapping through rigorous coding and testing to deliver high-quality software.
- Developed a responsive, accessible interface using Next.js and TypeScript, adhering to industry-wide technology trends and best practices for modern web development.

### AquaGuard: Real-Time Municipal Outbreak Surveillance & Forecasting System | Node.js, Express,

Aug 2025 - Oct 2025

Python, Time-Series Forecasting, JWT, Leaflet, Leaflet/Geospatial APIs

- Engineered the end-to-end full-stack platform for municipal health departments to monitor and forecast water-borne disease outbreaks, delivering high-quality software and sophisticated analytics.
- Developed interactive maps for real-time visualization of risk zones, demonstrating exceptional problem-solving ability by enabling officials to issue live public health alerts to stakeholders.
- Engineered a secure backend using Node.js and Express, implementing JWT Authentication and database-level protection to ensure Application Security and the integrity of sensitive epidemiological data.
- Optimized machine learning pipelines for anomaly detection in water quality parameters, achieving 88% prediction accuracy through rigorous coding and testing.

### NeuroPath: AI-Driven Behavioral Analysis & ASD Screening System | Scikit-learn, Seaborn,

Jun 2025 - Jul 2025

Imbalanced-learn, SMOTE, XGBoost, Pickle

- Developed the end-to-end a predictive model to identify the likelihood of Autism Spectrum Disorder (ASD), leveraging artificial intelligence to contribute to an inclusive and innovative culture.
- Built a robust analytics and machine learning pipeline, including data cleaning, EDA, and SMOTE for class imbalance, demonstrating a strong understanding of industry-wide technology trends.
- Iterated on the design and coding of multiple models (Decision Tree, Random Forest, XGBoost), achieving 82% accuracy through comprehensive testing and model tuning.
- Utilized developmental toolsets like Scikit-learn and Pickle to ensure the model was optimized for deployment and future software improvement.

## Technical Skills

**Languages:** Python, Java (5+ yrs), C, C++, JavaScript (ES6+), TypeScript, Bash, SQL

**AI/ML & Agentic AI:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LlamaIndex, LangGraph, Pandas, NumPy, Matplotlib, Seaborn

**Web & Full-Stack:** FastAPI, Next.js 15, React, Node.js, Express, Tailwind CSS, REST APIs, OpenAPI (Swagger)

**Databases & Vector Stores:** Big Data, Relational Databases (PostgreSQL), FAISS, ChromaDB, Pickle

**Developer Tools:** Git, GitHub, Maven, Docker, Linux, Postman, Vercel, Jupyter Notebook

**Key Concepts:** Agentic AI & Multi-Agent Systems, RAG (Retrieval-Augmented Generation), Model Context Protocol (MCP), Deep Learning (Transformers, CNNs), Natural Language Processing (NLP), Object-Oriented Programming (OOP), Agile (Scrum), CI/CD, SDLC, Unit Testing.